

SIVAKORN KHUNDILOKRATTAYA

Computer Engineering Student • Aspiring Data Scientist & ML Engineer

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EDUCATION

Srinakharinwirot University (SWU) <i>Bachelor of Engineering in Computer Engineering</i>	2024 – Present (Expected 2026) <i>Bangkok, Thailand</i>
Specialization in Statistical Machine Learning, Financial Risk Modeling, and Distributed Data Engineering.	

TECHNICAL SKILLS & COMPETENCIES

Languages: Python (Expert), SQL (PostgreSQL, MySQL), C, C++, Java, R, Bash scripting.
ML & Explainable AI: Scikit-Learn, LightGBM, XGBoost, SHAP (TreeExplainer), LIME, OptBinning, SMOTE-Tomek, Pandas, NumPy.
Data Engineering & Tools: PostgreSQL, MySQL, Apache Spark, Apache Airflow, Docker, FastAPI, Git.
Spoken Languages: Thai (Native), English (B1+ Professional), Japanese (Elementary).

FEATURED ENGINEERING PROJECTS

Enterprise Credit Risk Scoring & Explainable AI (XAI) <i>Python, LightGBM, SHAP, FastAPI</i> <i>Production-Grade Basel II/III Credit Decisioning Engine</i>	2026
- Engineered Weight of Evidence (WoE) and Information Value (IV) transformation pipeline to discretize continuous features into monotonic risk bins. - Integrated SHAP TreeExplainer to compute exact local Shapley values, automatically generating legally defensible Adverse Action notices (ECOA/FCRA compliant). - Deployed an asynchronous containerized FastAPI microservice delivering Probability of Default (PD) scoring with sub-50ms inference latency.	
Long Overdue Debtor (LOD) Classification System <i>LightGBM, XGBoost, Scikit-Learn</i> <i>Large-Scale Delinquency Prediction on ~40,000 Consumer Credit Records • Aihack 2025 Finalist</i>	Dec 2025
- Formulated temporal delinquency velocity ratios and revolving limit exhaustion slopes across 32,524 train / 8,619 test blind split samples. - Conducted 5-Fold Stratified Cross-Validation on tuned LightGBM and XGBoost ensembles, achieving 0.894 ROC-AUC and placing in the National Finalist tier.	
Industrial IoT & Manufacturing (MFG) Telemetry Pipeline <i>PostgreSQL, Spark, Airflow, Docker</i> <i>Automated Batch ETL & Real-Time Overall Equipment Effectiveness (OEE)</i>	2026
- Built multi-container telemetry pipeline ingesting high-frequency vibration, thermal, and cycle sensors across virtual factory lines. - Automated scheduled batch ETL in Airflow with stateful anomaly thresholding, surfacing real-time OEE tracking in interactive dashboards.	
GuardianAI – Clinical Frailty & Fall Risk Intervention <i>XGBoost, SHAP, FastAPI</i> <i>Dual-Layer Clinical Risk Prediction for Geriatric Patient Care • True Innovation Launchpad</i>	Jan 2026
Trained calibrated XGBoost models coupled with SHAP local biometric explainability for medical professionals, streaming through FastAPI.	

COMPETITIVE HACKATHONS & INDUSTRY CHALLENGES

AIRA & AIFUL / Aihack Thailand 2025 National Finalist

Dec 2025

Chulalongkorn Business School × ProbSpace

Bangkok, Thailand

- **Role & Contribution:** Machine Learning & Financial Risk Modeling Lead on ~40,000 real-world consumer loan samples.
- **Technical Rigor:** Built leak-free temporal feature matrices, handling class imbalance (1:12 default ratio) with SMOTE-Tomek and target encoding.
- **Executive Pitch:** Solved Question Sheet balancing default risk reduction against interest margin profitability, pitching to C-level executives.

True Innovation Launchpad 2026 – AI & Data Science Lead

Jan 2026

True Corporation Healthcare Innovation Track (Team Nakphatthana Tuapralat)

Bangkok, Thailand

- Spearheaded data strategy and predictive modeling for elderly frailty prevention, developing sub-30ms clinical risk scoring APIs.

Geospatial Intelligence for Resilience Hackathon 2026 – Spatial Lead

Jan 2026

GISTDA × KMITL × KMUTT Spatial Data Scientist & ML Modeler

Bangkok, Thailand

- Built "Rain-to-Flood" spatial model fusing Sentinel-1 SAR soil backscatter, Copernicus GLO-30 DEM, and precipitation APIs for 30m flash flood mapping.

LINE MAN Wongnai Junior Case Competition – Strategic Growth Lead

Nov 2024

LINE MAN Wongnai Strategic Growth & Unit Economics Track (Team LC)

Bangkok, Thailand

- Formulated campus delivery optimization and "Silent Mission" growth campaign with 8.87 THB/user CAC, acquiring 462 VIP trial conversions.

INDUSTRY CERTIFICATIONS & STANDARDIZED ASSESSMENTS

Data Analyst Assessment (STAMP) – Score: 97/150 (Intermediate)

March 2026

SKILLKAMP by KBTG (Kasikorn Business-Technology Group) • Benchmark: 90

- **Core Scores:** Data Cleaning & Preprocessing: 23.00/30 | EDA & Viz: 20.00/30 | ML for Analysts: 18.00/30 | Foundations: 20.00/30

Digital and Performance Marketer Assessment (STAMP) – Score: 98/150 (Intermediate)

March 2026

SKILLKAMP by KBTG • Benchmark: 94

- **Top Performer:** Digital Marketing Analytics: 26.00/30 (Benchmark: 17.94) | Data-Driven Marketing: 21.00/30 | Content: 19.00/30.

Oxford Placement Test – Score: 51/120 (CEFR B1+ Professional)

June 2026

AUA Language Center / Oxford University Press

- Grammar & Use of English: 50.88 (B1) | Listening Proficiency: 51.12 (B1) – Verified professional working proficiency in English.

COURSEWORK & SOFTWARE ENGINEERING DISCIPLINE

Core Coursework: Machine Learning, Database Systems, Operating Systems, Statistical Methods for Engineers, Software Engineering, Data Structures & Algorithms, Computer Networks.

Engineering Discipline: Adheres to PEP-8 style standards, modular OOP pipeline design, reproducible containerized Docker environments, and zero black-box deployments without local feature attribution.